Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

Anonymous ACL submission

Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy's entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token's update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. We show the blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term's prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We then derive an estimator for the visitation channel that costs one extra forward pass. The reward-to-go implied by the decomposition is the remaining conditional entropy H_{togo}, which we forecast with parallel multi-token-prediction heads; the sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout in which several samples share a prefix and diverge at scheduled depths. The resulting advantage A_H sums to zero within a sibling set by construction, so the method cannot change the average amount of damping — it can only decide which of two siblings is damped. We give the full derivation, the four approximations it requires and what each discards, and a five-arm design that isolates the contribution of the forecast's *value* from that of the apparatus carrying it.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model's output distribution. Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token's update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x. The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token's surprisal from the mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.4). Meanwhile the visitation term's prefactor is $1 - \pi_\theta(y_t \mid s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo}; subtracting any function of the state leaves it unbiased. We estimate H_{togo} with parallel multi-token-prediction (MTP) heads read off the hidden state the policy already computes, and we obtain the state-only baseline from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. "Total entropy went up" is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Section 9.3).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.4).
- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo}, together with an explicit ledger of the approximations that estimation requires, the sign of each one's error, and the mechanism that absorbs it (Sections 5–10).
- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A five-arm experimental design that separates the contribution of the forecast's value from the contribution of the apparatus that delivers it, by running the full pipeline twice with the forecast's value destroyed in two different ways (Section 11).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

> **Assumption 1** (parameter-independent softmax; Hao et al., 2026). For any state s, each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token's update induces (Hao et al., 2026). That work observes that earlier interventions are "indirect" and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025d) show that restricting policy-gradient updates to a small minority of high-entropy "forking" tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Wu et al. (2025a) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Section 6 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

3 Preliminaries

Notation. A prompt is x, a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot \mid s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a \mid s_t)$, so that $H = \mathbb{E}_{a\sim\pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \mathrm{mean}_j r_j)/\mathrm{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token's contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token's update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our 2), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$
\begin{aligned}
\Omega(s) = -\frac{\eta}{L}\,\mathbb{E}_a\Big[&\frac{\mathbb{I}_{\mathrm{clip}}(s,a)\,A(s,a)}{\pi_{\mathrm{old}}(a \mid s)} \\
&\cdot \pi_\theta(a \mid s)\big(1-\pi_\theta(a \mid s)\big) \\
&\cdot \big(\log \pi_\theta(a \mid s) + H\big)\Big],
\end{aligned} \tag{1}
$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\mathrm{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$
\alpha_u = \exp\!\Big(-\alpha\,\frac{|\Omega_u|}{\max_v |\Omega_v|}\Big), \qquad \lambda_{\min} = e^{-\alpha}, \tag{2}
$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\mathrm{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Section 8).

4 Two Channels

4.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$
\mathcal{H}(\pi_\theta \mid x) := -\sum_y \pi_\theta(y \mid x)\,\log \pi_\theta(y \mid x), \tag{3}
$$

a sum over an exponentially large set. The autoregressive factorization $\pi_\theta(y \mid x) = \prod_t \pi_\theta(y_t \mid s_t)$ turns the logarithm into a sum and lets us exchange the order of summation:

$$
\begin{aligned}
\mathcal{H} &= -\sum_y \pi_\theta(y \mid x) \sum_{t=1}^{T} \log \pi_\theta(y_t \mid s_t) \\
&= \sum_{t=1}^{T} \mathbb{E}_{y\sim\pi_\theta}\big[-\log \pi_\theta(y_t \mid s_t)\big].
\end{aligned} \tag{4}
$$

Each summand is still an expectation over whole responses, but the integrand depends on y only through $y_{\le t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\mathbb{E}_y\big[-\log \pi_\theta(y_t \mid s_t)\big] = \mathbb{E}_{y_{<t}}\Big[\underbrace{\mathbb{E}_{y_t\sim\pi_\theta(\cdot|s_t)}\big[-\log \pi_\theta(y_t \mid s_t)\big]}_{=\,H(\pi_\theta(\cdot|s_t))}\Big],$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t. Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_\theta \mid x) = \mathbb{E}_{y\sim\pi_\theta}\Big[\sum_{t=1}^{T} H\big(\pi_\theta(\cdot \mid s_t)\big)\Big]. \qquad (5)$$

Equation (5) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_\theta(\cdot \mid s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (5) that is easy to pass over: *the measure $\mathbb{E}_{y\sim\pi_\theta}$ depends on θ as well.* The right-hand side is not a sum of θ-dependent terms over a fixed set of positions — it is a θ-dependent average of them over a θ-dependent set of trajectories.

4.2 Step 2: the product rule

Write the per-trajectory total

$$f_\theta(y) := \sum_{t=1}^{T} H\big(\pi_\theta(\cdot \mid s_t(y))\big), \qquad (6)$$

so that Eq. (5) reads $\mathcal{H} = \sum_y \pi_\theta(y \mid x)\, f_\theta(y)$. Both factors of every summand carry θ, so differentiating requires the product rule:

$$\nabla_\theta \mathcal{H} = \sum_y \big[\nabla_\theta \pi_\theta(y \mid x)\big] f_\theta(y) + \sum_y \pi_\theta(y \mid x)\, \nabla_\theta f_\theta(y). \qquad (7)$$

The second sum is already an expectation. The first is not, because $\nabla_\theta \pi_\theta$ is not a probability distribution; the score-function identity $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ restores it to one. Substituting and naming the two terms,

$$\nabla_\theta \mathcal{H} = \underbrace{\mathbb{E}\big[\nabla_\theta f_\theta(y)\big]}_{\text{(L) local}} + \underbrace{\mathbb{E}\big[f_\theta(y)\, \nabla_\theta \log \pi_\theta(y \mid x)\big]}_{\text{(V) visitation}}. \qquad (8)$$

Equations (3)–(8) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

4.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the sub-tree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

4.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level. Collecting the constants of Eq. (1) into $c := \eta\, \mathbb{I}_{\text{clip}} A/(L\, \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a,

$$\Omega \;\propto\; c\, \pi_a(1-\pi_a)\, \big(H - I_a\big), \qquad (9)$$

and $H = \mathbb{E}_{a\sim\pi}[I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot \mid s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a, so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability —

are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer. This is the step that closes the argument, and it follows from Hao et al.'s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_\theta \mid s)}{\partial z_{s,a}} = -\,\pi_\theta(a \mid s)\big(\log \pi_\theta(a \mid s) + H(\pi_\theta \mid s)\big), \tag{10}$$

and at a uniform distribution $\log \pi_\theta(a \mid s) = -\log k = -H$ makes the bracket vanish for *every* a. So $\partial H/\partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^\delta + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^\delta}{S}, \qquad H''(0) = -\frac{k-1}{k^2}, \tag{11}$$

with $H'(\delta)$ of sign $-\operatorname{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4}, roughly two orders of magnitude below that of a peaked distribution under the same perturbation. *As far as local entropy is concerned, uniform branch points are genuinely safe*, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term's prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than "the local term is incomplete": *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

distribution	H	$P(\text{split})$	$\mathbb{E}\lvert\Omega\rvert/\lvert A\rvert$	$\mathbb{E}[1-\pi_a]$
q=.99	0.10	0.08	0.092	0.020
q=.9, tail	1.25	0.57	**1.129**	0.190
q=.6, m=3	1.11	0.98	0.457	0.587
unif. k=10	2.30	**1.00**	**0.000**	**0.900**
unif. k=100	4.61	**1.00**	**0.000**	**0.990**

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q, the remaining $1-q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that n=8 samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (8) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (6) and the score by $\nabla_\theta \log \pi_\theta(y \mid x) = \sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)$ — turns (V) into a double sum:

$$(\mathrm{V}) = \sum_t \sum_u \mathbb{E}\big[H(\pi_\theta(\cdot \mid s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u \mid s_u)\big]. \tag{12}$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot \mid s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned}\mathbb{E}\big[\nabla_\theta \log \pi_\theta(y_u \mid s_u) \mid y_{<u}\big] &= \sum_a \pi_\theta(a \mid s_u) \frac{\nabla_\theta \pi_\theta(a \mid s_u)}{\pi_\theta(a \mid s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a \mid s_u) \;=\; \nabla_\theta 1 \;=\; 0.\end{aligned} \tag{13}$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u, the entropies at all later positions:

$$(\mathrm{V}) = \mathbb{E}\Big[\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)\, H_{\text{togo}}(s_u \oplus y_u)\Big], \tag{14}$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$.

We stress that H_{togo} was not posited. Equation (14) is a REINFORCE estimator whose reward is "the conditional entropy remaining after this token", and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (14) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \tag{15}$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\le u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO's group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout's own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Four Approximations

Equation (15) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k\, H\big(\pi(\cdot \mid s_{u+k})\big). \tag{16}$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ.

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (15) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: forecasting the unrealized future

Even Eq. (16) requires the future states s_{u+k}. At update time they do not exist: rollouts came from π_{old}, each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u:

$$H\big(\pi(\cdot \mid s_{u+k})\big) \;\longrightarrow\; H\big(p^{(k)}_{\text{MTP}}(\cdot \mid h_u)\big). \tag{17}$$

Head k is a two-layer residual MLP applied to h_u, followed by the policy's own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy's own next-token distribution and the forecast starts from a sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u, the k-th term of Eq. (16) has expectation $H(Y_{u+k} \mid Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} \mid S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} \mid S_u) - \mathbb{E}\big[H(\pi(\cdot \mid S_{u+k-1}))\big] \\ = I\big(Y_{u+k}; Y_{u+1:u+k-1} \mid S_u\big) \\ \ge 0. \end{aligned} \tag{18}$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k.

Equation (18) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

6.3 A3: per-head affine calibration

Equation (18) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k\, H\big(p^{(k)}_{\mathrm{MTP}}(\cdot \mid h_u)\big) + b_k. \tag{19}$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (18) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \geq 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\hat{H}_{\mathrm{togo}}(s_u \oplus y_u) = \max\Big(0,\ \sum_{k=1}^{\kappa} \gamma_H^k \big(a_k H_k(h_u) + b_k\big)\Big), \tag{20}$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\mathrm{togo}} = 0.72\, H_1 + 0.06\, H_2 + 0.11$: the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (18) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u; it does not know y_u. The quantity we need, $H_{\mathrm{togo}}(s_u \oplus y_u)$, must therefore be read from position $u + 1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

7 From Gradient to Token Weights

Equation (15) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u, changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \mathrm{clip}(r_u, 1 - \epsilon_{\mathrm{lo}}, 1 + \epsilon_{\mathrm{hi}})A_u$ the coefficient the policy loss actually applies, Assumption 2 makes a single step move only the sampled token's logit, $\Delta z_{u,a} = \eta\, w_u\, \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi_u} = \eta\, w_u\, (1 - \pi_u), \tag{21}$$

which vanishes as $\pi_u \to 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_\theta \log \pi_u \cdot \Delta\theta = \Delta \log \pi_u$, substituting into Eq. (15) gives the per-token quantity

$$\mathrm{visit}_u := \widehat{\Delta \log \pi_u} \cdot \mathrm{clip}\big(A_H(s_u, y_u), -c, +c\big). \tag{22}$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token's influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u.

Sign quadrants. Equation (22) defines the collapse channel precisely. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi_u}$ reverses both readings.

8 Combining the Channels

The two channels enter the same weight:

$$\tilde{\Omega}_u = \mathrm{norm}(\Omega_u) + \lambda \cdot \mathrm{norm}(\mathrm{visit}_u), \tag{23}$$

after which $\tilde{\Omega}$ is passed to the *unmodified* exponential map of Eq. (2), producing $\alpha_u \in [\lambda_{\min}, 1]$. We emphasize that the hook is inserted at the single point where Ω is finalized and that nothing downstream is changed; when $\lambda = 0$ the pipeline is bitwise identical to the base method, which we enforce with a regression test.

Three choices in Eq. (23) deserve comment. Normalization is required even though both terms carry units of nats, because Ω contains a factor $1/\pi_{\mathrm{old}}$ and is heavy-tailed. Because normalization removes scale, the step size η in Eq. (21) only ever appears multiplied by λ; we fix $\eta = 1$ and sweep λ alone. Finally, norm is an RMS rescaling rather than a z-score, since subtracting a mean would break the $\lambda = 0$ equivalence.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{\le u}$ produce identical forecasts. Combined with Eq. (15) this gives a sharp characterization:

$$A_H[i,u] \neq 0 \iff \begin{array}{l}\text{siblings share } y_{<u} \\ \text{but diverge at } y_u,\end{array} \tag{24}$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (24) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (24) it counts branch points, and n sequences form a trie with at most $n-1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n-1)/T$ for *any* sampler whatsoever. With $n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree's role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \text{sib}(s_u)} A_H\big(s_u, y_u^{(j)}\big) = 0. \tag{25}$$

The correction in Eq. (23) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. "Mean entropy at branch points increased" is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (8).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 18), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

11 Experimental Setup

11.1 Model, data, and optimization

All runs fine-tune the same base model on the same data with the same optimizer settings, and differ only in the intervention. We use a 1.5B-parameter mathematics-specialized model, train on a 17k-problem competition-mathematics set, and validate on AIME 2024 (30 problems) with 32 samples per problem. With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs.

Optimization uses GRPO with $n = 8$ responses per prompt, a constant learning rate of 1e−6 with no warm-up (so that the schedule is independent of the declared total step count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. Rollouts are sampled at temperature 1.0 with no nucleus truncation; validation uses temperature 1.0 and top-p = 0.7. Every arm uses the same data seed and the same initial checkpoint, which

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (5)–(15)	nothing — identities	—	—
A1	horizon κ, discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\to$ MTP marginal	mutual information with intervening tokens, Eq. (18)	**over-estimate, monotone in k**	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	*inherited from the base method*
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

we verify by requiring the step-0 validation scores to agree.

11.2 Method hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (19). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method's exponential map of Eq. (2) with $\lambda_{\min} = 0.7$, unchanged and unrescaled, so weights land in $[0.7, 1]$ and can only attenuate.[1] The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves.

11.3 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that carries it? A comparison between the base method and the full method changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H.

UNIFORM applies the same attenuation to every position in the support, discarding A_H's value but keeping its support — this is the implementation of the "raise the mean" hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (25) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
STEER-F	.25	tree	A_H as derived

Table 3: The five arms. UNIFORM and PERMUTED run the full pipeline with the forecast's value destroyed in two different ways, so any gain they show is attributable to the apparatus rather than to the visitation signal.

11.4 Metrics

We report acc@32 (mean accuracy over 32 samples) and maj@32 (accuracy of the majority vote over the same 32 samples), and their difference

$$\text{uplift} := \text{maj@32} - \text{acc@32},$$

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than pass@32 because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, SE =

[1] Our implementation exposes an optional reshaping of $\tilde{\Omega}$ before this map (winsorizing or rank-transforming it); every run reported here uses the identity setting, so the base method's map sees exactly what it would have seen.

step	GRPO	STEER	UNIF	PERM	**STEER-F**
0	.039/.035	.043/.040	.039/.035	.039/.035	.039/.035
10	.075/.127	.064/.109	.070/.124	.062/.126	.078/.148
20	.120/.176	.114/.195	.102/.171	.087/.155	.103/.181
30	.108/.195	.135/.215	.121/.223	.099/.178	.113/.211
40	.144/.236	.115/.187	.119/.201	.116/.197	.141/.230
50	.120/.204	.125/.196	.121/.216	.153/.224	.145/.258
60	.126/.178	.139/.202	.125/.203	.135/.190	.145/.237
70	.130/.200	.128/.200	.129/.197	.132/.227	.147/.257
80	.135/.182	.135/.202	.140/.204	.134/.197	.144/.206
90	.136/.191	.144/.212	.142/.178	.130/.210	**.166**/.239
100	.132/.171	.146/.207	.134/.192	.144/.212	.157/.222
110	.157/.208	.132/.181	.140/.171	.152/.218	.151/.230

Table 4: AIME24 acc@32/maj@32 at every validation point. Rows below the rule are the converged window used for all subsequent statistics.

$\text{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged `actor/entropy` is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity and, where the distinction could matter, the length-corrected product as well.

12 Results

All numbers below come from the runs described in Section 11: one seed, one benchmark, 110 optimization steps per arm, and the same initial checkpoint. We report our own measurements only; no figure in this section is taken from prior work.

12.1 Validation trajectory

Table 4 gives AIME24 accuracy and majority-vote accuracy at every validation point. The four arms that share the tree pipeline start from an identical step-0 score; STEER starts at .043 rather than .039, a gap we attribute to re-sampling noise rather than to a different checkpoint, since two validations of the *same* base weights in our logs differ by .004 in accuracy.

12.2 Converged window

Table 5 averages each arm over steps 40–110.

arm	acc@32	maj@32	uplift	entropy	len
GRPO	.1350	.1963	.0612	**.1491**	964
STEER	.1330	.1984	.0654	.1214	988
UNIFORM	.1313	.1953	.0640	.1374	946
PERMUTED	.1370	.2094	.0724	.1190	960
STEER-F	**.1495**	**.2349**	**.0854**	.1210	984

Table 5: Means over the converged window (steps 40–110, eight validation points per arm). *uplift* is maj@32 − acc@32; *len* is the mean response length in tokens. Entropy is the token-mean quantity discussed in Section 11, i.e. $\mathcal{H}$ divided by mean length.

contrast	Δacc	t	Δmaj	t
STEER-F − GRPO	+.0145	3.08	+.0386	4.74
STEER-F − STEER	+.0165	6.68	+.0365	5.09
STEER-F − UNIF	+.0182	7.18	+.0396	5.50
STEER-F − PERM	+.0125	2.56	+.0255	5.26
GRPO − STEER	+.0020	0.34	−.0021	−0.21
PERM − STEER	+.0040	0.83	+.0110	1.75
UNIF − STEER	−.0017	−0.62	−.0031	−0.52

Table 6: Paired contrasts over the eight converged-window steps. t is the paired t-statistic on those eight differences; it asks whether a gap is consistent across the window, not whether it would survive a change of benchmark or seed.

Two orderings are worth stating before the contrasts. On accuracy, the two arms with no forecast (STEER, UNIFORM) sit *below* plain GRPO, and only the arm carrying A_H is clearly above it. On entropy the ordering is almost reversed: GRPO retains the most entropy and is second-to-last on accuracy, while STEER-F retains among the least and is first.

12.3 Does the forecast's value matter?

Table 6 pairs the arms step by step over the converged window and reports the mean difference together with a paired t over the eight points.

The block structure is the result. Every contrast involving A_H separates; no contrast among the arms without it does. In particular UNIFORM runs the entire apparatus — tree sampler, forecaster, support, weight map — and differs from STEER by $-.0017$ ($t = -0.62$). Turning the machinery on buys nothing. PERMUTED keeps the machinery *and* the magnitudes of A_H and destroys only the pairing between a branch and its score; it recovers about a quarter of the gain ($+.0040$ of $+.0165$) and the rest requires the pairing to be correct.

This is the contrast the design was built for. A comparison of STEER-F against STEER alone

contrast	Δentropy	Δacc
GRPO − STEER	+.0277	+.0020
GRPO − STEER-F	+.0281	−.0145
STEER − STEER-F	+.0004	−.0165

Table 7: Aggregate entropy does not order the arms. The last row is the decisive one: two arms whose converged entropy differs by .0004 differ in accuracy by .0165 and in majority vote by .0365.

changes two things at once, and cannot separate the forecast from the sampler that delivers it; the two destroyed-forecast arms do separate them.

12.4 Where the gain comes from

The majority-vote gain decomposes into a part explained by per-sample accuracy and a part explained by the sample set's structure:

$$\underbrace{+.0386}_{\Delta\text{maj}} = \underbrace{+.0145}_{\Delta\text{acc (38\%)}} + \underbrace{+.0241}_{\Delta\text{uplift (62\%)}},$$

against GRPO, with the uplift term at $t = 4.82$; against STEER the uplift term is $+.0200$ ($t = 3.26$). Nearly two thirds of the improvement is not that individual samples are more often right, but that the right ones agree with each other more and the wrong ones agree less. Since UNIFORM shows no uplift gain, the structural change is attributable to A_H and not to the tree.

12.5 Entropy is not the channel

Table 7 makes three separate points. First, both intervened arms hold *less* entropy than plain GRPO, so whatever they are doing, it is not suppressing the collapse. Second, the arm with the most entropy is not the arm with the most accuracy. Third, and decisively, STEER and STEER-F converge to the same entropy to within .0004 — the same number of paths remain open — and yet differ by .0165 in accuracy. The quantity that separates them is not how much entropy survives but which branches it survives on.

This is what Section 9.3 predicted. A_H sums to zero over a sibling set, so it cannot move the mean damping; measured over the whole run the mean token weight is 0.999 in every arm. An intervention that is mean-preserving by construction *should* leave the entropy trajectory alone, and it does.

12.6 Mechanism diagnostics

Section 9 bounds the fraction of positions that can carry a non-zero A_H at $2(n-1)/T$ for any sampler. With $n = 8$ and a mean response length of 984 tokens the bound is .0142. Measured over training, the three tree arms sit at .0136 (UNIFORM), .0129 (PERMUTED) and .0120 (STEER-F), i.e. at 84–96% of a ceiling that no sampler can exceed. A plain-rollout run with the same λ reaches .003.

Two readings follow. The tree sampler is close to saturating what the combinatorics allow, so there is little headroom in scheduling more cuts; and the intervention is extremely sparse — about one position in eighty — which makes the size of the effect in Table 6 the surprising part rather than the small support.

12.7 Cost

Within the runs executed on identical hardware, STEER takes 833 s per optimization step and STEER-F takes 1516 s, an overhead of 82%. The UNIFORM and PERMUTED arms cost 1480 and 1492 s, confirming that the cost is the tree rollout plus the MTP forward pass and not the A_H computation, which is a subtraction over a mask. We do not report a GRPO timing because that arm ran on a different machine.

12.8 A note on how these numbers were selected

We report window means rather than the best checkpoint, and the difference is not cosmetic. Selecting the maximum over the twelve validation points inflates every arm by a similar amount: +.013 for STEER, +.011 for UNIFORM, +.016 for PERMUTED and +.017 for STEER-F, relative to the window mean. Since a single AIME24 measurement carries a standard error of .029 across problems, a maximum over a dozen noisy points is a biased statistic whose bias grows with the number of checkpoints compared. The arm ordering happens to be the same under both rules here, but we regard the window mean as the defensible one and use it throughout.

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. We showed that the omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. We then showed that the omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, that a parallel forecaster estimates it for the price of one forward pass with an error whose sign and monotonicity are known in closed form, and that a tree-structured rollout supplies the baseline it requires. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it.

Limitations

A single validation benchmark, and what our t-statistics do not cover. Every claim rests on AIME 2024, which has 30 problems. A single measurement of one arm carries an across-problem standard error of .029, so the gaps in Table 6 are a fraction of one problem in absolute terms. The paired t we report is computed over the eight validation steps of one run: it establishes that a gap is stable across the converged window, and nothing more. The statistic that would speak to generalization — the across-problem standard error of the *paired* difference, in which per-problem difficulty cancels between two arms evaluated on the same 30 questions — is not recoverable from our logs, which record per-benchmark means and standard deviations rather than per-problem scores. We state this rather than substitute a number we cannot compute. A multi-benchmark evaluation, and logging per-problem scores so the paired error can be estimated, are the two things that would settle it.

One row of our tables is not yet reproducible from the artifact. The GRPO row comes from the same experiment campaign, hardware and protocol as the others, but its training log is not in the repository we will release, whereas every other row is recomputed from a committed log. It should be treated as the one number a reader cannot yet check.

A single seed. All arms use one data seed. The standard error we quote is an across-problem quantity and does not bound run-to-run variation, which for these runs is uncontrolled because rollout sampling is not seeded across processes. A re-run of the same configuration is a second draw, not a restoration.

The effective horizon is one step. The calibration in Eq. (19) suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is dominated by a single step ahead. This is the predicted behavior of Eq. (18) rather than a bug, but it means the "future" in the visitation term is a short future, and a forecaster whose error did not grow with k would test the idea more sharply. A sequential forecaster is the obvious comparison and we have not run it.

The support is small by construction. Section 9 bounds the fraction of positions carrying a non-zero A_H at $2(n-1)/T$, on the order of a percent for our settings. Whether the effect scales with n, or with a branching schedule chosen adaptively rather than at fixed depths, is untested.

An assumption with no remedy. Approximation B2 — that a token's update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method rather than introduced here, but our construction adds a second estimator that relies on it.

One prediction of the analysis is unmeasured. Section 4.4 argues that the two channels are anti-correlated. The prefactor half of that claim is analytic and the A_H half is inferred from a correlation with branch diversity; we have not measured their *product* at real sibling-divergence positions in a live rollout. Until that measurement exists, Table 1 is a prediction rather than an observation.

Scale. All experiments use a single 1.5B model. The relative size of the two channels plausibly depends on how sharp the base policy already is, and we have no evidence about larger models.

Ethics Statement

This work studies an optimization-time regularizer for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our method inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the method (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Section 12.7 so that it can be weighed against the benefit.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, , and Sara Hooker. 2024. Back to basics: Revisiting reinforce-style optimization for learning from human feedback in llms. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 12248–12267.

Shengnan An, Xunliang Cai, Xuezhi Cao, Xiaoyu Li, Yehao Lin, Junlin Liu, Xinxuan Lv, Dan Ma, Xuanlin Wang, Ziwen Wang, , et al. 2025. Amo-bench: Large language models still struggle in high school math competitions. ArXiv preprint arXiv:2510.26768.

Anonymous. 2025. Rethinking entropy regularization in large reasoning models. ArXiv preprint arXiv:2509.25133.

Anthropic. 2025. Introducing claude opus 4.5. Https://www.anthropic.com/news/claude-opus-4-5.

Sikai Bai, Haoxi Li, Jie Zhang, Yongjiang Liu, , and Song Guo. 2026. Ttvs: Boosting self-exploring reinforcement learning via test-time variational synthesis. ArXiv preprint arXiv:2604.08468.

Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwei Peng, Jason D. Lee, Deming Chen, and Tri Dao. 2024. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. ArXiv preprint arXiv:2401.10774.

Yupeng Chang, Yi Chang, , and Yuan Wu. 2026. Balora: Bias-alleviating low-rank adaptation to mitigate catastrophic inheritance in large language models. In The Fourteenth International Conference on Learning Representations.

Yupeng Chang, Chenlu Guo, Yi Chang, , and Yuan Wu. 2025. Lora-mgpo: Mitigating double descent in low-rank adaptation via momentum-guided perturbation optimization. in findings of the association for computational linguistics: Emnlp 2025, pages 648–659.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, , et al. 2024. A survey on evaluation of large language models. acm transactions on intelligent systems and technology, 15(3):1–45.

Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, , and Xiangnan He. 2023. Bias and debias in recommender system: A survey and future directions. acm transactions on information systems, 41(3):1–39.

Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Wayne Xin Zhao, Zhenliang Zhang, , and Furu Wei. 2025. Reasoning with exploration: An entropy perspective. ArXiv preprint arXiv:2506.14758.

Xiangxiang Chu, Hailang Huang, Xiao Zhang, Fei Wei, , and Yong Wang. 2025. Gpg: A simple and strong reinforcement learning baseline for model reasoning. ArXiv preprint arXiv:2504.02546.

Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, , et al. 2025a. Process reinforcement through implicit rewards. ArXiv preprint arXiv:2502.01456.

Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, , et al. 2025b. The entropy mechanism of reinforcement learning for reasoning language models. ArXiv preprint arXiv:2505.22617.

Yu Cui, Feng Liu, Jiawei Chen, Canghong Jin, Xingyu Lou, Changwang Zhang, Jun Wang, Yuegang Sun, , and Can Wang. 2025c. Hatllm: Hierarchical attention masking for enhanced collaborative modeling in llm-based recommendation. ArXiv preprint arXiv:2510.10955.

Jia Deng, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, , and Ji-Rong Wen. 2025. Decomposing the entropy-performance exchange: The missing keys to unlocking effective reinforcement learning. ArXiv preprint arXiv:2508.02260.

Chongming Gao, Kexin Huang, Jiawei Chen, Yuan Zhang, Biao Li, Peng Jiang, Shiqi Wang, Zhong Zhang, , and Xiangnan He. 2023a. Alleviating matthew effect of offline reinforcement learning in interactive recommendation. In Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval, pages 238–248.

Chongming Gao, Shiqi Wang, Shijun Li, Jiawei Chen, Xiangnan He, Wenqiang Lei, Biao Li, Yuan Zhang, , and Peng Jiang. 2023b. Cirs: Bursting filter bubbles by counterfactual interactive recommender system. acm transactions on information systems, 42(1):1–27.

Google. 2025. A new era of intelligence with gemini 3. Https://blog.google/products/gemini/ gemini-3/.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, , et al. 2024. The llama 3 herd of models. ArXiv preprint arXiv:2407.21783.

Aaron Grattafiori et al. 2024. The llama 3 herd of models. ArXiv preprint arXiv:2407.21783.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, , et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. ArXiv preprint arXiv:2501.12948.

Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, , and Sergey Levine. 2018. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. in international conference on machine learning, pages 1861–1870. pmlr.

Yaru Hao, Li Dong, Xun Wu, Shaohan Huang, Zewen Chi, , and Furu Wei. 2025. On-policy rl with optimal reward baseline. ArXiv preprint arXiv:2505.23585.

Zhezheng Hao, Hong Wang, Haoyang Liu, Jian Luo, Jiarui Yu, Hande Dong, Qiang Lin, Can Wang, and Jiawei Chen. 2026. Rethinking entropy interventions in RLVR: An entropy change perspective. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 31105–31133.

Andre He, Daniel Fried, , and Sean Welleck. 2025a. Rewarding the unlikely: Lifting grpo beyond distribution sharpening. ArXiv preprint arXiv:2506.02355.

Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, , et al. 2024. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. ArXiv preprint arXiv:2402.14008.

Jujie He, Jiacai Liu, Chris Yuhao Liu, Rui Yan, Chaojie Wang, Peng Cheng, Xiaoyu Zhang, Fuxiang Zhang, Jiacheng Xu, Wei Shen, , et al. 2025b. Skywork open reasoner 1 technical report. ArXiv preprint arXiv:2505.22312.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, , and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the math dataset. ArXiv preprint arXiv:2103.03874.

Jian Hu, Jason Klein Liu, Haotian Xu, , and Wei Shen. 2025a. Reinforce++: An efficient rlhf algorithm with robustness to both prompt and reward models. ArXiv preprint arXiv:2501.03262, 1(3):5.

Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang, Xiangyu Zhang, , and Heung-Yeung Shum. 2025b. Open-reasoner-zero: An open source approach to scaling up reinforcement learning on the base model. ArXiv preprint arXiv:2503.24290.

Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, , et al. 2024. Qwen2.5-coder technical report. ArXiv preprint arXiv:2409.12186.

Zed Industries. 2025. zeta. Https://huggingface.co/datasets/zed-industries/zeta.

Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, , et al. 2024. Openai o1 system card. ArXiv preprint arXiv:2412.16720.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, , and Ion Stoica. 2024. Livecodebench: Holistic and contamination free evaluation of large language models for code. ArXiv preprint arXiv:2403.07974.

Albert Q. Jiang et al. 2023. Mistral 7b. ArXiv preprint arXiv:2310.06825.

Yihang Jiang, Xiaoyang Li, Guangxu Zhu, Hang Li, Jing Deng, Kaifeng Han, Chao Shen, Qingjiang Shi, , and Rui Zhang. 2023. 6g non-terrestrial networks enabled low-altitude economy: Opportunities and challenges. ArXiv preprint arXiv:2311.09047.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, , et al. 2024. Tulu 3: Pushing frontiers in open language model post-training. ArXiv preprint arXiv:2411.15124.

Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, , et al. 2022. Solving quantitative reasoning problems with language models. advances in neural information processing systems, 35:3843–3857.

Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q Jiang, Ziju Shen, , et al. 2024. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions. hugging face repository, 13(9):9.

Mengdi Li, Jiaye Lin, Xufeng Zhao, Wenhao Lu, Peilin Zhao, Stefan Wermter, , and Di Wang. 2025a. Curriculum-rlaif: Curriculum alignment with reinforcement learning from ai feedback. ArXiv preprint arXiv:2505.20075.

Qingbin Li, Rongkun Xue, Jie Wang, Ming Zhou, Zhi Li, Xiaofeng Ji, Yongqi Wang, Miao Liu, Zheming Yang, Minghui Qiu, , et al. 2025b. Cure: Critical-token-guided re-concatenation for entropy-collapse prevention. ArXiv preprint arXiv:2508.11016.

Siyi Lin, Chongming Gao, Jiawei Chen, Sheng Zhou, Binbin Hu, Yan Feng, Chun Chen, , and Can Wang. 2025. How do recommendation models amplify popularity bias? an analysis from the spectral perspective. In Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, pages 659–668.

Aixin Liu, Aoxue Mei, Bangcai Lin, Bing Xue, Bingxuan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang, Chaofan Lin, Chen Dong, , et al. 2025a. Deepseek-v3. 2: Pushing the frontier of open large language models. ArXiv preprint arXiv:2512.02556.

Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, , and Yi Dong. 2025b. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. ArXiv preprint arXiv:2505.24864.

Ziyu Liu, Yuhang Zang, Shengyuan Ding, Yuhang Cao, Xiaoyi Dong, Haodong Duan, Dahua Lin, , and Jiaqi Wang. 2025c. Spark: Synergistic policy and reward co-evolving framework. ArXiv preprint arXiv:2509.22624.

Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi, William Y Tang, Manan Roongta, Colin Cai, Jeffrey Luo, Tianjun Zhang, Li Erran Li, , et al. 2025. Deepscaler: Surpassing o1-preview with a 1.5 b model by scaling rl. notion blog.

Shichao Ma, Zhiyuan Ma, Ming Yang, Xiaofan Li, Xing Wu, Jintao Du, Yu Cheng, Weiqiang Wang, Qiliang Liu, Zhengyang Zhou, , et al. 2026. Tspo: Breaking the double homogenization dilemma in multi-turn search policy optimization. ArXiv preprint arXiv:2601.22776.

Volodymyr Mnih, Adria Puigdomenech Badia, Mehdi Mirza, Alex Graves, Timothy Lillicrap, Tim Harley, David Silver, , and Koray Kavukcuoglu. 2016. Asynchronous methods for deep reinforcement learning. in international conference on machine learning, pages 1928–1937. pmlr.

OpenAI. 2025. Introducing gpt-5.2. Https://openai.com/index/introducing-gpt-5-2.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, , and Oleg Klimov. 2017. Proximal policy optimization algorithms. ArXiv preprint arXiv:1707.06347.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, , et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. ArXiv preprint arXiv:2402.03300.

Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, , and Chuan Wu. 2025. Hybridflow: A flexible and efficient rlhf framework. In Proceedings of the Twentieth European Conference on Computer Systems, pages 1279–1297.

Yuda Song, Julia Kempe, , and Remi Munos. 2025. Outcome-based exploration for llm reasoning. ArXiv preprint arXiv:2509.06941.

Hongze Tan and Jianfei Pan. 2025. Gtpo and grpo-s: Token and sequence-level reward shaping with policy entropy. ArXiv preprint arXiv:2508.04349.

Haozhe Wang, Qixin Xu, Che Liu, Junhong Wu, Fangzhen Lin, , and Wenhu Chen. 2025a. Emergent hierarchical reasoning in llms through reinforcement learning. ArXiv preprint arXiv:2509.03646.

Hong Wang, Zhezheng Hao, Jian Luo, Chenxing Wei, Yao Shu, Lei Liu, Qiang Lin, Hande Dong, , and Jiawei Chen. 2025b. Scheduling your llm reinforcement learning with reasoning trees. ArXiv preprint arXiv:2510.24832.

Jiakang Wang, Runze Liu, Fuzheng Zhang, Xiu Li, , and Guorui Zhou. 2025c. Stabilizing knowledge, promoting reasoning: Dual-token constraints for rlvr. ArXiv preprint arXiv:2507.15778.

Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, , et al. 2025d. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. ArXiv preprint arXiv:2506.01939.

Ziqing Wang, Yibo Wen, William Pattie, Xiao Luo, Weimin Wu, Jerry Yao-Chieh Hu, Abhishek Pandey, Han Liu, , and Kaize Ding. 2025e. Polo: Preferenceguided multi-turn reinforcement learning for lead optimization. ArXiv preprint arXiv:2509.21737.

Fang Wu, Weihao Xuan, Ximing Lu, Zaid Harchaoui, , and Yejin Choi. 2025a. The invisible leash: Why rlvr may not escape its origin. ArXiv preprint arXiv:2507.14843.

Fei Wu, Zhenrong Zhang, Qikai Chang, Jianshu Zhang, Quan Liu, , and Jun Du. 2026. Step potential advantage estimation: Harnessing intermediate confidence and correctness for efficient mathematical reasoning. ArXiv preprint arXiv:2601.03823.

Junkang Wu, Kexin Huang, Jiancan Wu, An Zhang, Xiang Wang, , and Xiangnan He. 2025b. Quantile advantage estimation for entropy-safe reasoning. ArXiv preprint arXiv:2509.22611.

Zhenhe Wu, Jian Yang, Jiaheng Liu, Xianjie Wu, Changzai Pan, Jie Zhang, Yu Zhao, Shuangyong Song, Yongxiang Li, , and Zhoujun Li. 2025c. Tabler1: Region-based reinforcement learning for table understanding. ArXiv preprint arXiv:2505.12415.

Can Xie, Ruotong Pan, Xiangyu Wu, Yunfei Zhang, Jiayi Fu, Tingting Gao, , and Guorui Zhou. 2025. Unlocking exploration in rlvr: Uncertainty-aware advantage shaping for deeper reasoning. ArXiv preprint arXiv:2510.10649.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, , et al. 2024a. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. ArXiv preprint arXiv:2409.12122.

Shihui Yang, Chengfeng Dou, Peidong Guo, Kai Lu, Qiang Ju, Fei Deng, , and Rihui Xin. 2025a. Dcpo: Dynamic clipping policy optimization. ArXiv preprint arXiv:2509.02333.

Weiqin Yang, Jiawei Chen, Xin Xin, Sheng Zhou, Binbin Hu, Yan Feng, Chun Chen, , and Can Wang. 2024b. Psl: Rethinking and improving softmax loss from pairwise perspective for recommendation. advances in neural information processing systems, 37:120974–121006.

Weiqin Yang, Jiawei Chen, Shengjia Zhang, Peng Wu, Yuegang Sun, Yan Feng, Chun Chen, , and Can Wang. 2025b. Breaking the top-k barrier: Advancing top-k ranking metrics optimization in recommender systems. In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2, pages 3542–3552.

Weiqin Yang, Bohao Wang, Zhenxiang Xu, Jiawei Chen, Shengjia Zhang, Jingbang Chen, Canghong Jin, , and Can Wang. 2026. Bear: Towards beam-searchaware optimization for recommendation with large language models. ArXiv preprint arXiv:2601.22925.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, , et al. 2025. Dapo: An open-source llm reinforcement learning system at scale. ArXiv preprint arXiv:2503.14476.

Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, , and Junxian He. 2025. Simplerlzoo: Investigating and taming zero reinforcement learning for open base models in the wild. ArXiv preprint arXiv:2503.18892.

Kaiyan Zhang, Yuxin Zuo, Bingxiang He, Youbang Sun, Runze Liu, Che Jiang, Yuchen Fan, Kai Tian, Guoli Jia, Pengfei Li, , et al. 2025a. A survey of reinforcement learning for large reasoning models. ArXiv preprint arXiv:2509.08827.

Kangning Zhang, Wenxiang Jiao, Kounianhua Du, Yuan Lu, Weiwen Liu, Weinan Zhang, , and Yong Yu. 2025b. Looptool: Closing the data-training loop for robust llm tool calls. ArXiv preprint arXiv:2511.09148.

Ruipeng Zhang, Ya-Chien Chang, , and Sicun Gao. 2025c. When maximum entropy misleads policy optimization. ArXiv preprint arXiv:2506.05615.

Shengjia Zhang, Weiqin Yang, Jiawei Chen, Peng Wu, Yuegang Sun, Gang Wang, Qihao Shi, , and Can Wang. 2026a. Talos: Optimizing top-k accuracy in recommender systems. ArXiv preprint arXiv:2601.19276.

Wenyuan Zhang, Xinghua Zhang, Haiyang Yu, Shuaiyi Nie, Bingli Wu, Juwei Yue, Tingwen Liu, , and Yongbin Li. 2026b. Expseek: Self-triggered experience seeking for web agents. Preprint, arXiv:2601.08605.

Xingjian Zhang, Siwei Wen, Wenjun Wu, , and Lei Huang. 2025d. Edge-grpo: Entropy-driven grpo with guided error correction for advantage diversity. ArXiv preprint arXiv:2507.21848.

Xinglang Zhang, Yunyao Zhang, ZeLiang Chen, Junqing Yu, Wei Yang, , and Zikai Song. 2026c. Logical phase transitions: Understanding collapse in llm logical reasoning. Preprint, arXiv:2601.02902.

Yunyao Zhang, Xinglang Zhang, Junxi Sheng, Wenbing Li, Junqing Yu, Yi-Ping Phoebe Chen, Wei Yang, , and Zikai Song. 2026d. Semantic-aware logical reasoning via a semiotic framework. Preprint, arXiv:2509.24765.

Ziqi Zhao, Zhaochun Ren, Jiahong Zou, Liu Yang, Zhiwei Xu, Xuri Ge, Zhumin Chen, Xinyu Ma, Daiting Shi, Shuaiqiang Wang, , et al. 2026. Reinforced efficient reasoning via semantically diverse exploration. ArXiv preprint arXiv:2601.05053.

Yixiao Zhou, Yang Li, Dongzhou Cheng, Hehe Fan, , and Yu Cheng. 2026. Look inward to explore outward: Learning temperature policy from llm internal states via hierarchical rl. ArXiv preprint arXiv:2602.13035.

Xinyu Zhu, Mengzhou Xia, Zhepei Wei, Wei-Lin Chen, Danqi Chen, , and Yu Meng. 2025. The surprising effectiveness of negative reinforcement in llm reasoning. ArXiv preprint arXiv:2506.01347.

Brian D Ziebart, Andrew Maas, J Andrew Bagnell, , and Anind K Dey. 2008. Maximum entropy inverse reinforcement learning. In AAAI, volume 8, pages 1433–1438.

A Closed Forms for Section 4.4

Perturbing one logit of a k-way uniform distribution by δ and writing $S := e^{\delta} + k - 1$ gives $\pi_1 = e^{\delta}/S$ and $\pi_j = 1/S$ for $j \neq 1$, so

$$H(\delta) = \log S - \frac{\delta e^{\delta}}{S},$$
$$H'(\delta) = -\frac{(k-1)\,\delta\, e^{\delta}}{S^2},$$

whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$. The sign of H' is $-\operatorname{sign}(\delta)$, so any perturbation lowers the entropy, but only at second order. Table 8 contrasts this with a peaked distribution, where the first-order term does not vanish.

B Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\mathrm{eq}[i,j,t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t, so that $\mathrm{prefixeq}[i,j,t]$ is true exactly when the two rollouts agree on $y_{\leq t}$.

distribution	H_0	Ω	ΔH at δ=.1
uniform, k=10	2.303	0	-4.7×10^{-4}
uniform, k=50	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	-0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	-0.396	-3.8×10^{-2}

Table 8: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

Shifting it one position right gives agreement on $y_{<t}$, which is the sibling relation Eq. (15) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

Alignment. $\hat{H}_{\text{togo}}$ is read at position $t+1$ and moved to position t; at the final position the value is kept. Omitting this shift scores the state before the branch instead of the branch taken.

Memory. The RL path never materializes head logits. For each head the hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding, reduced to an entropy, and released, so peak memory is $O(\text{chunk} \times V)$ rather than $O(\kappa \times B \times T \times V)$.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method's on random inputs; the $\lambda = 0$ early return in the combination step is what makes this hold exactly rather than to floating-point tolerance.